HARISH KUMAR R

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SUMMARY

A motivated Computer Science graduate with strong analytical skills and a solid understanding of programming and AI fundamentals. Eager to learn, adapt, and apply knowledge to real-world engineering problems

Experience

Freelance AI / Computer Vision Engineer
Self-Employed | Jan-2026 – Mar-2026

- Designed and developed **computer vision solutions** for real-world client use cases using **deep learning** and **image processing** techniques
- Built and trained **custom object detection models** (YOLO) for symbol and pattern recognition tasks
- Performed **data collection, annotation, preprocessing**, and dataset balancing to improve model accuracy

AI Engineer Intern
Global Tech Professionals (United Kingdom) | Jun 2026 – Present
(Remote)

- Developed AI and computer vision solutions for client projects using Python, OpenCV, and deep learning techniques.
- Built, trained, and optimized YOLO-based object detection models for real-world image analysis and recognition tasks.
- Collected, annotated, and preprocessed datasets, applying augmentation and balancing techniques to improve model performance.
- Collaborated with a remote engineering team to develop, test, and deploy AI models while following software development best practices**.**

TECHNICAL SKILLS

- **Programming Languages:** Python
- **Machine Learning**: PyTorch, NumPy, Pandas, Scikitlearn, Matplotlib
- **Framework:** FastAPI,Langchain
- **Computer Vision:** YOLO,OpenCV
- **Tools:** Github,Docker
- **Database:** MySQL

CERTIFICATIONS

- Certified in Python Programming(Infosys SpringBoard)
- Certified in Big Data Computing(NPTEL)
- Certified in HTML,CSS

EDUCATION

Erode Sengunthar Engineering College
M.Tech – Computer Science and Engineering (Integrated)
CGPA: 7.5 | 2022 – Present | Erode, Tamil Nadu

PROJECTS

ATTENDANCE SYSTEM / Nov-2025

Tech Stack: FastAPI

- Developed an attendance system using FastAPI.
- Sends SMS notifications to students' parents for attendance updates.

Blueprint Symbol Detection and Estimation /Dec-2025 - Jan 2026
Tech Stack: YOLOv8, Python, OpenCV, Deep Learning

- Built an AI system to detect and count engineering symbols from blueprint drawings.
- Applied image preprocessing techniques to improve detection accuracy.
- Converted blueprint drawings into structured, machine-readable data

BUDDY – AI Interviewer / Feb-2026
Tech Stack:: HTML, CSS, Python, Groq LLM API

- Built an AI-powered mock interview platform to simulate real interview conversations.
- Integrated LLM inference using Groq API to generate interview questions and evaluate responses.
- Developed a web interface using HTML and CSS for interactive interview practice.
- Implemented Python backend to process user inputs and return AI-generated feedback.

Electricity Demand Forecasting / Mar-2026 – April-2026
Tech Stack: Python, Pandas, NumPy, Scikit-learn

- Built a machine learning model to predict electricity demand using historical and weather data
- Performed data preprocessing and feature engineering (time-based and lag features)
- Evaluated model performance using RMSE (Baseline: 195, Final: 250)
- Improved usability with user-friendly input handling and modifications

Fingerprint-Based Blood Group Prediction / Apr-2026 – Present
Tech Stack: Python, PyTorch, Streamlit, OpenCV

- Built a Convolutional Neural Network (CNN) model to classify blood groups using fingerprint images
- Performed image preprocessing and trained a multi-class classification model for 8 blood group types
- Developed an interactive Streamlit web app for real-time prediction with confidence scores

WoundCare-AI | Intelligent Wound Assessment & Monitoring System
Backend & AI Engineer | Group Project | Mar 2026 – Present
Tech Stack: YOLOv8, EfficientNet ,Flutter, FastAPI, RabbitMQ, PostgreSQL, Docker, OpenCV,

- Developed YOLOv8-based wound detection and segmentation models for automated wound analysis from medical images.
- Built an EfficientNet classification model and integrated AI inference services using FastAPI and RabbitMQ.
- Collaborated in developing a scalable wound monitoring platform with PostgreSQL, Docker, and microservice-based architecture.

YouTube AI-Powered Tutor | Jul-2026 – Present
Tech Stack: Python, Streamlit, LangChain, FAISS, HuggingFace Embeddings, Groq API, YouTube Transcript API

- Developed an AI-powered tutoring application that answers user questions based on YouTube video transcripts using a Retrieval-Augmented Generation (RAG) pipeline.
- Implemented transcript extraction, semantic search with FAISS and HuggingFace embeddings, and integrated the Groq LLM to generate context-aware responses.
- Built an interactive Streamlit web application enabling users to input YouTube URLs, process transcripts, and receive accurate real-time answers through a user-friendly interface